

PROJECT REPORT

IP6.a

Modernization of a Legacy 3D Printer System

Printer Model	BIBO 2 Touch
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Abstract

This report documents the analysis, planning, and ongoing modernization of a BIBO dual-extruder FDM 3D printer as part of the IP6.a interdisciplinary student project. The printer was found to be non-functional at the start of the project, and the first phase of work focused on restoring basic operation through firmware configuration, with several technical problems encountered and documented.

The hardware modernization plan addresses the key limitations of the BIBO system: dual-extruder complexity will be reduced to a single-extruder configuration to improve reliability and reduce moving mass; the unstable two-part bed assembly will be replaced with a modern spring-steel magnetic PEI sheet system; thermal insulation will be added to the heated bed; and a door-locking mechanism will be fitted. Internal LED lighting is also planned to improve print visibility. The software component, led by a second team member, addresses Klipper firmware integration and web-based remote control.

A structured before-and-after performance testing methodology has been defined: the printer will be tested with a standard benchmark part before and after the hardware upgrades to quantify improvement. This report covers the full literature review, system analysis, upgrade plan, problems encountered, and project timeline.

1. Introduction

This project focuses on the comprehensive modernization of a BIBO dual-extruder FDM 3D printer. The BIBO is a commercially produced, enclosed desktop printer featuring a rigid aluminum frame, a red acrylic enclosure, a dual-extruder system, and a heated bed. While the hardware design is generally solid, the printer was found to be non-functional at the start of this project, and its current configuration presents several practical limitations that impair print quality and usability.

The project is conducted by a two-person student team under the IP6.a interdisciplinary framework. One student is responsible for the hardware and mechanical/electrical upgrades; the second student is responsible for the software, firmware configuration, and any computer-side integration. The project combines elements of mechanical engineering, embedded systems, and software development.

The modernization strategy proceeds in two major phases. In the first phase, the printer is restored to working condition through firmware configuration and basic calibration. In the second phase, targeted hardware upgrades are implemented. A structured performance test using an identical benchmark part before and after the upgrades provides a quantitative basis for evaluating the impact of the changes.

This report presents the technical background (literature review), a detailed analysis of the existing BIBO system including its component inventory, a full account of the configuration challenges encountered, the planned upgrade strategy, and the project timeline.

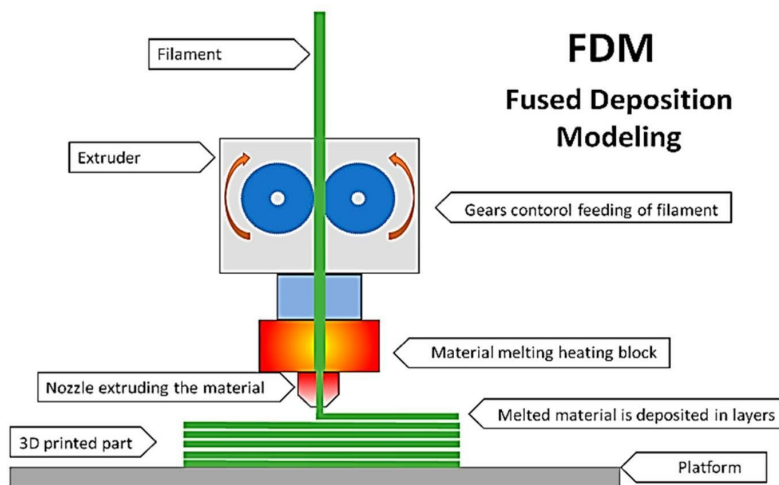
2. Literature Review

This chapter reviews the technical foundations relevant to the project. The goal is to establish the background knowledge that informs the design decisions made throughout the modernization process.

2.1 3D Printing Technologies

Fused Deposition Modeling (FDM)

FDM is the dominant desktop 3D printing technology and the process used by the BIBO printer. A thermoplastic filament is fed through a heated nozzle (the hot end) and extruded onto a build platform layer by layer until the part is complete. Key quality factors include: layer height, print speed, nozzle temperature, bed temperature, cooling, and the precision of the motion system.



Dual-Extruder FDM

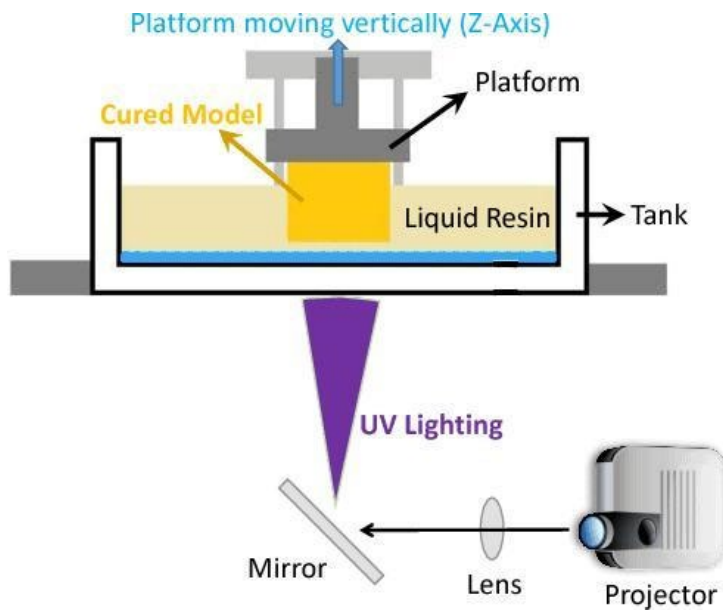
Some FDM printers, including the BIBO, feature two independent extruders and hot ends on the same carriage. This enables multi-material or dual-colour printing. However, dual extruders have significant drawbacks: the increased carriage mass reduces achievable print speed and acceleration; the second extruder occupies build volume, reducing the effective maximum part size; and offset calibration between the two nozzles is complex and a frequent source of print defects. For this project, the decision has been made to remove the second extruder to reduce mass and simplify the system.

Lithography-Based Methods (SLA)

Lithography-based 3D printing methods, such as Stereolithography (SLA), use ultraviolet light to cure liquid photopolymer resin layer by layer. Unlike FDM systems that extrude molten plastic, SLA printers selectively harden resin using a laser or LCD masking system.

SLA printing provides significantly higher surface quality and dimensional precision compared to FDM printing. It is widely used for dental applications, prototyping, jewelry production, and detailed miniature models. However, SLA systems generally require more expensive materials, additional post-processing, and careful handling of chemical resins.

Compared to FDM, SLA printers typically achieve finer layer resolution and smoother surfaces, but they often have smaller build volumes and higher operational costs. FDM remains more suitable for functional mechanical parts, while SLA is preferred for high-detail visual models.



2.2 Kinematic Architectures

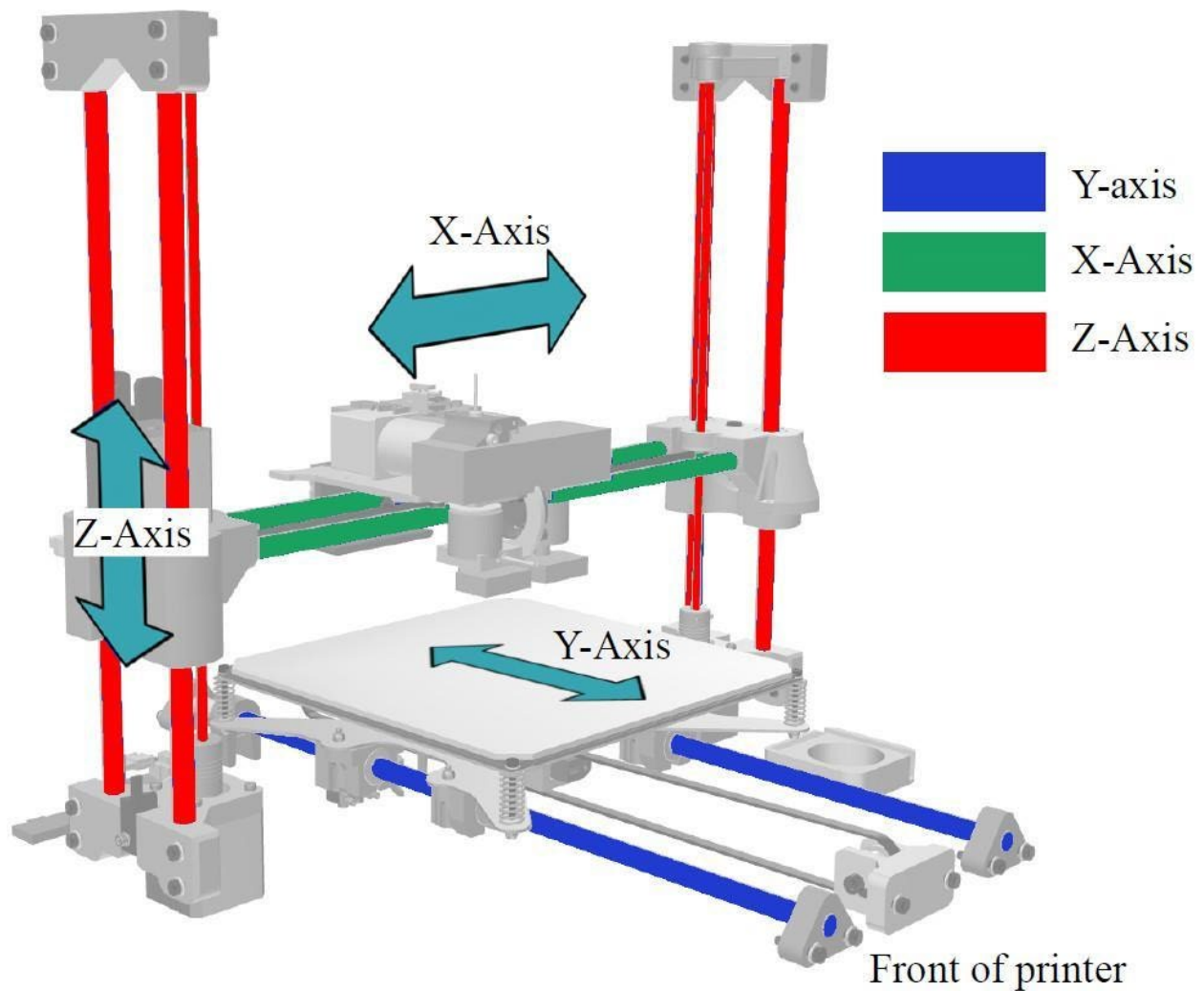
The BIBO uses a Cartesian kinematic architecture, in which the X, Y, and Z axes are driven independently. The print head moves in X, the bed moves in Y, and the Z axis raises the X-axis assembly. This is a well-understood, mechanically simple architecture, but the moving bed creates inertia in the Y direction that limits print speed.

Feature	Cartesian (BIBO)	CoreXY	Delta
Bed movement	Moves in Y	Stationary	Stationary
Max speed	Medium	High	Very High
Complexity	Low	Medium	High
Calibration	Straightforward	Moderate	Complex

Cartesian Systems

Cartesian 3D printers use independent X, Y, and Z axes arranged orthogonally. Each motor controls motion along a single axis, making the system mechanically simple and easy to understand. Cartesian systems are widely used in consumer printers because of their reliability, low cost, and ease of maintenance.

The primary advantages of Cartesian systems include simple kinematics, straightforward firmware configuration, and accurate positioning. However, moving-bed designs can introduce vibration and inertia at higher speeds, reducing print quality during fast movements.



In our printer the bed moves in the z axis and the head moves in the X and Y axis

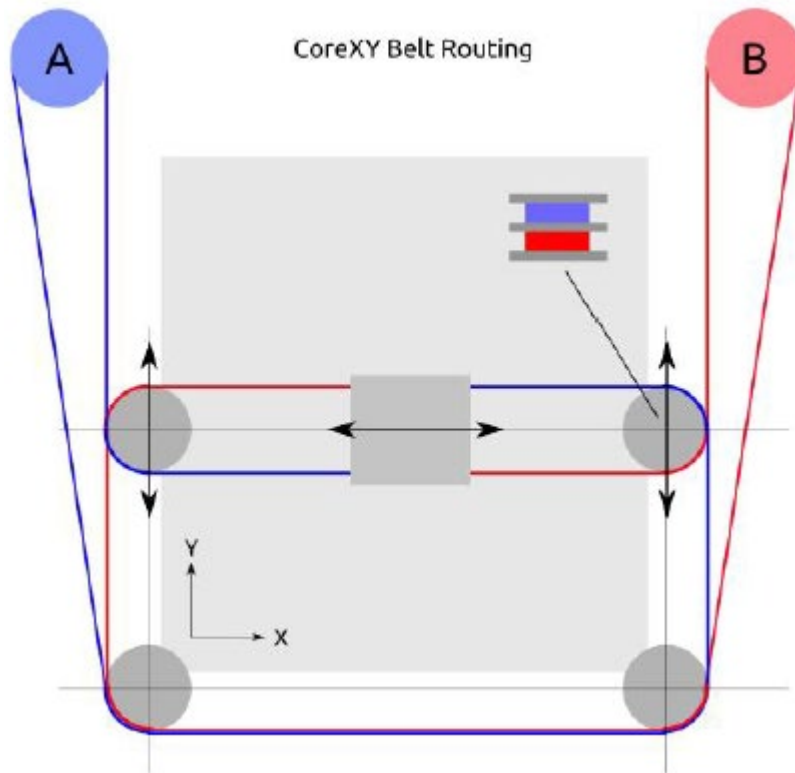
CoreXY Systems

CoreXY printers use a belt-driven motion system where two motors work together to control X and Y movement. This design reduces moving mass because the motors remain fixed to the frame, allowing faster acceleration and improved high-speed printing performance.

CoreXY architectures provide better speed-to-quality performance than many Cartesian systems.

They are commonly used in modern high-speed printers because they reduce vibration and improve dynamic motion behavior. However, the belt routing and calibration process are more complex, increasing setup difficulty and maintenance requirements.

Klipper firmware is particularly popular in CoreXY systems due to its advanced motion planning and resonance compensation features.



Delta Systems

Delta printers use three vertically mounted towers connected to the print head through parallel arms. Coordinated movement of the towers determines the nozzle position. Delta systems are capable of extremely fast motion and smooth movement because the print head has very low moving mass. The main advantages of Delta printers are high speed and smooth curved motion. However, calibration and firmware configuration are significantly more complex than Cartesian systems. Delta printers also often have reduced usable print volume near the edges of the build area. Because of their mathematical complexity, Delta systems benefit greatly from modern firmware with advanced kinematic calculations and automatic calibration tools.

2.3 Open-Source Ecosystem and the RepRap Project

The RepRap project, initiated by Adrian Bowyer in 2005, established the foundational conventions of modern open-source FDM printing: G-code control, modular hardware design, and community-driven firmware. The BIBO printer, like most consumer FDM printers, is descended from this ecosystem and ships with old Klipper Version.

2.4 Firmware: Marlin vs. Klipper

Marlin

Marlin runs entirely on the printer's microcontroller and handles all motion planning, G-code parsing, and peripheral control in real time. It is the standard firmware shipped with the BIBO and most consumer FDM printers. Its limitations include constrained computational resources, and the requirement to recompile and re-flash the firmware for any configuration change.

Klipper

Klipper offloads computationally intensive tasks (motion planning, kinematics, G-code processing) to a host computer (a Raspberry Pi), leaving only time-critical step-pulse generation on the microcontroller. Configuration is done via a human-readable text file; no recompilation is required. Klipper supports advanced features such as input shaping (resonance compensation via IMU) and pressure advance. The Moonraker API server and Mainsail web frontend provide a full web-based control interface.

Aspect	Marlin (current)	Klipper (target)
Computation	Microcontroller only	Linux PC+ MCU
Reconfiguration	Recompile & flash	Edit config file, restart
Web interface	None	Mainsail / Fluidd via Moonraker
Advanced features	Limited	Input shaping, pressure advance

2.5 Mechanical Design Principles

Relevant mechanical design principles (Slocum, 1992) that apply to this project include:

- Frame rigidity: deformation under load introduces positional error.
- Moving mass minimization: lighter carriages allow higher acceleration without ringing.
- Thermal management: uniform bed temperature is critical for part adhesion. Silicone spacers and insulation boards under the heated bed improve thermal stability and reduce heat loss downward.
- Vibration and resonance: printer frames act as resonant structures. Damping feet and structural stiffening reduce amplitude.

2.6 Calibration and Sensor Systems

Bed Leveling

The print bed must be coplanar with the nozzle's travel plane to a fraction of the first layer height. Automatic bed leveling (ABL) using a probe (e.g., BLTouch or CR Touch) maps the bed surface and compensates for tilt and warp in firmware. Klipper supports full bed mesh compensation.

Resonance Compensation

At high print speeds, frame and toolhead vibrations produce ringing artifacts in the printed part. Klipper's input shaping feature uses an ADXL345 accelerometer mounted on the toolhead to measure resonant frequencies and applies a filter to cancel them.

Bed Thermal Insulation

Heated beds lose a significant fraction of their electrical energy as heat conducted downward into the frame. Fitting a thermal insulation pad (typically a silicone or cork sheet) between the bed heater and the frame reduces warm-up time, improves temperature stability, and lowers power consumption. This is a recommended upgrade for the BIBO system.

3. Printer Overview: BIBO System Analysis

3.1 BIBO Printer Specifications

The printer being modernized is a BIBO dual-extruder FDM desktop printer. It is an enclosed Cartesian-kinematics printer with a rigid aluminum frame and red acrylic side and top panels. Key specifications are summarized below.

Manufacturer	BIBO 2 Touch
Kinematics	Cartesian (bed moves in Y, toolhead in X, Z lifts X-axis)
Extruder type	Dual extruder (Bowden-style feed)
Firmware (stock)	Klipper
Frame	Aluminum extrusion with acrylic enclosure panels
Enclosure	Red acrylic top lid and side panels, hinged front door
Display	Small LCD/touchscreen panel on front-right
Target firmware	Klipper (via Linux PC)

3.2 Component Inventory

The following table lists the major components identified in the BIBO printer. Components marked with an asterisk (*) are planned for replacement or modification as part of this project.

Component	Description / Specification	Status
Frame	Aluminum extrusion + acrylic panels	Keep
X-axis carriage	Dual hot-end carriage on linear rods	Modify (remove 2nd extruder)
Hot end (×2)	Proprietary BIBO hot ends with heater block and nozzle	1 removed, 1 kept
Extruder motor (×2)	NEMA 17 stepper, drives filament feed	1 removed, 1 kept
X-axis motor	NEMA 17 stepper	Keep
Y-axis motor	NEMA 17 stepper	Keep
Z-axis motor(s)	NEMA 17 stepper(s)	Keep
Heated bed	Two-part assembly: aluminium base + glass top layer with clamps	Replaced (PEI magnetic system)
Bed insulation	None currently fitted	Added (silicone/cork pad)
Control board	8-bit or 32-bit controller	Upgraded
Power supply	Internal PSU	Keep
Endstops	Mechanical or optical endstops on X, Y, Z	Keep / verify
Belts & pulleys	GT2 belts, 20-tooth pulleys on X and Y	Inspect and replace if worn
Z leadscrew(s)	Leadscrew + brass nut for Z-axis	Keep

Display / interface	Front panel with buttons or touchscreen	Replace
Acrylic door	Front hinged access door	Add locking mechanism
Front door latch	None currently fitted	Add (3D-printed PETG or ABS part)
Internal lighting	Small led from front(not enough)	Add (LED strip around the whole internal structure)
Cooling fan(s)	Hotend cooling fan + part cooling fan(s)	1 removed, 1 kept
Damping/rubber legs	Stock rigid frame(no legs)	Replace (3D-printed TPU 95A parts for vibration isolation)
Plastic extruder	Stock plastic extruder	Replace with metal extruder

3.3 Initial Assessment and Identified Limitations

The following limitations were identified during the initial assessment of the printer:

- **Non-functional state:** The printer could not be operated at the start of the project and required a full firmware re-configuration before any testing could be done.
- **Dual extruder complexity:** The second extruder adds mass to the carriage and reduces the maximum printable part size. Because both nozzles cannot be at the same Z height simultaneously, parts are limited in size, and nozzle offset calibration is a persistent source of error.
- **Unstable bed assembly:** The heated bed consists of a lower aluminium plate (the original bed, screwed to the frame) and a glass sheet placed on top and held in position with small metal clamps. This arrangement is mechanically unstable, difficult to level reliably, and was identified as the cause of several ruined prints where parts bonded to the glass surface and could not be removed cleanly. The glass surface scratches over time and becomes progressively less effective.
- **No bed thermal insulation:** The heated bed currently has no insulation underneath, resulting in slow heat-up times and poor temperature uniformity.
- **Door self-closing:** The front acrylic door has no latch or holding mechanism and closes on its own due to any vibration or movement during printing, making it difficult to observe the print or intervene without accidentally disturbing the printer.
- **No internal lighting:** The interior of the printer is poorly lit, making it difficult to observe the print process or identify early failure signs.

4. Configuration Process and Problems Encountered

4.1 Configuration Approach

Since the printer was not operational at the start of the project, the first task was to restore it to a working state. The team followed the Klipper installation and configuration procedure documented at. The process involved the following steps:

1. Install Klipper, Moonraker, and Mainsail using KIAUH (Klipper Installation And Update Helper).
2. Compile and flash the Klipper MCU firmware onto the printer's control board.
3. Write an initial printer.cfg configuration file for the BIBO, defining kinematics, axis parameters, stepper motor settings, endstops, heater pins, thermistor types, and fans.
4. Connect the PC to the printer's control board via USB.
5. Verify communication and attempt a first homing sequence.

The configuration of a non-standard printer such as the BIBO requires careful identification of all pin assignments on the control board, which are not always publicly documented. This phase proved more time-consuming than anticipated due to several technical problems described below.

The installation of Klipper, Moonraker, and Mainsail was performed using KIAUH (Klipper Installation And Update Helper), a script-based installer that automates the deployment of the full Klipper software stack on a PC. While KIAUH simplifies the individual installation of each component, significant difficulties were encountered in establishing reliable communication between them. The Moonraker API server, which acts as the bridge between the Klipper host process and the Mainsail web frontend, initially failed to connect to the Klipper Unix Domain Socket. This resulted in Mainsail reporting the printer as 'disconnected' despite Klipper appearing to run correctly.

4.2 Problems Faced

Each problem encountered during the configuration phase is documented in detail in the project repository, including the symptom observed, the diagnostic steps taken, and the resolution applied. This documentation serves both as a project record and as a reference for anyone attempting a similar configuration on a BIBO printer.

The following is a template for how each problem will be recorded:

Field	Content
Problem(1) title	End stop of nozzle
Symptom	The nozzle that we are using cant reach the end of the bed
Root cause	The fan of the nozzle we are not using is blocking the way
Resolution	Remove the fan
Reference	
Problem(2) title	Moonraker unable to connect to Klipper socket
Symptom	Mainsail displayed the printer as permanently "disconnected"; no printer state was visible despite both Klipper and Moonraker services running correctly
Root cause	The klippy_uds_address parameter in moonraker.conf contained the path /home/pi/printer_data/comms/klippy.sock, which is the default PC path. The actual system user is cpsstudend, so the socket was created at /home/cpsstudend/printer_data/comms/klippy.sock. Moonraker was therefore looking in a location that did not exist.

Resolution	Corrected the klippy_uds_address in moonraker.conf to /home/cpsstudend/printer_data/comms/klippy.sock and restarted Moonraker
Reference	https://moonraker.readthedocs.io/en/latest/configuration/
Problem(3) title	Mainsail macros not loaded (virtual_sdcard, pause_resume, etc.)
Symptom	Mainsail displayed multiple warnings: virtual_sdcard, pause_resume, gcode_macro PAUSE, gcode_macro RESUME, gcode_macro CANCEL_PRINT, and display_status not defined in config
Root cause	The [include mainsail.cfg] line was present in printer.cfg, but mainsail.cfg was a symlink pointing to mainsail-config/mainsail.cfg, which was itself a symlink pointing to ./client.cfg. The final target file client.cfg existed but was completely empty (0 bytes), so no macros were ever loaded
Resolution	Wrote the correct Mainsail client macro content directly into /home/cpsstudend/mainsail-config/client.cfg and performed a FIRMWARE_RESTART
Reference	
Problem(4) title	All endstops permanently reading TRIGGERED
Symptom	QUERY_ENDSTOPS returned TRIGGERED for all three axes (X, Y, Z) with nothing physically pressing any endstop. Homing failed immediately with "Endstop stepper_x still triggered after retract"
Root cause	The endstop pins in printer.cfg were defined with the ^ prefix (e.g. ^PE5), which enables the microcontroller's internal pull-up resistor. The BIBO's mechanical endstops are normally-closed (NC) type, meaning they read as triggered when the pull-up is active and nothing is pressing them — the logic is inverted
Resolution	Removed the ^ prefix from all three endstop pin definitions in printer.cfg (stepper_x, stepper_y, stepper_z) and performed a FIRMWARE_RESTART. All endstops then correctly reported open at rest
Reference	
Problem(5) title	MCU firmware version mismatch
Symptom	Mainsail displayed a warning: "MCU 'mcu' has deprecated code (it is missing feature 'STEPPER_STEP_BOTH_EDGE'). Recompiling and flashing is recommended." MCU was running v0.12.0-85 while the host was running v0.13.0-658
Root cause	The Klipper MCU firmware flashed onto the printer's ATmega2560 control board was outdated relative to the Klipper host version installed on the PC
Resolution	Recompiled Klipper firmware using make menuconfig (target: Atmel AVR, atmega2560, 250000 baud), then flashed it using make flash FLASH_DEVICE=/dev/serial/by-id/usb-1a86_USB2.0-Serial-if00-port0 and restarted Klipper
Reference	
Problem(6) title	Klipper running on a PC instead of Raspberry Pi

Symptom	Several default configuration paths and assumptions in Moonraker and Klipper documentation reference /home/pi/ and Raspberry Pi-specific tooling, none of which applied to this installation
Root cause	The project runs Klipper on a standard Ubuntu 24.04 desktop workstation (cps-wkstn-i71) rather than a Raspberry Pi. This caused path mismatches and meant that Pi-specific guidance had to be adapted throughout
Resolution	All paths were updated to reflect the actual user home directory /home/cpsstudend/. ModemManager (a desktop Linux service that can seize serial ports and block Klipper) was verified and disabled as a precaution
Reference	Moonraker docs: https://moonraker.readthedocs.io/en/latest/configuration/
Problem(7) title	under extrusion
Symptom	Printed parts showed gaps between extrusion lines, poor layer bonding, and visibly thin or missing material in areas that should be solid
Root cause	PID control parameters for the hotend were not tuned for the BIBO hardware, causing temperature fluctuations during printing. Unstable hotend temperature leads to inconsistent filament viscosity and flow rate, resulting in under-extrusion
Resolution	PID calibration was performed using Klipper's built-in PID_CALIBRATE command. The hotend was calibrated at the target print temperature by running PID_CALIBRATE HEATER=extruder TARGET=200 in the Mainsail console. Klipper automatically ran a heating cycle, calculated the optimal P, I, and D values, and the results were saved permanently to printer.cfg using SAVE_CONFIG. The same process was repeated for the heated bed using PID_CALIBRATE HEATER=heater_bed TARGET=60
Reference	Obico — Klipper PID Tuning Guide: https://www.obico.io/blog/klipper-pid-tuning/
Problem(8) title	Unexpected emergency stop triggered during active print
Symptom	Klipper triggers an emergency stop mid-print with a heater fault error. The print is interrupted and the printer requires a manual restart to resume operation.
Root cause	The extruder motor was faulty. It would randomly stall or stop during printing, triggering Klipper's safety mechanism. The issue was hardware-related, not a firmware or wiring problem.
Resolution	The faulty motor was replaced with the second extruder motor from the original dual-extruder setup. The new motor was installed and calibrated (rotation distance and PID tuning), and the printer has been working reliably since.
Reference	
Problem(9) title	Printed parts undersized
Symptom	Printed parts were consistently smaller than designed. A test square designed to be 20 mm measured only 16.5 mm with calipers

Root cause	The rotation_distance value in printer.cfg for the affected axis was set to the Klipper default and did not match the actual mechanical configuration of the BIBO printer (belt pitch, pulley tooth count, or leadscrew pitch). This parameter defines how many mm of movement result from one full motor rotation, so an incorrect value causes all movements to be scaled wrongly
Resolution	The correct rotation distance was calculated from the measured deviation using the formula: $\text{new_rotation_distance} = (\text{measured} / \text{expected}) \times \text{current_rotation_distance}$. With a measured dimension of 16.5 mm against an expected 20 mm, and a current value of 40: $(16.5 / 20) \times 40 = 33$. The rotation distance was updated to 33 in printer.cfg, and the test square was reprinted to confirm correct dimensions
Reference	Klipper Documentation — Rotation Distance: https://www.klipper3d.org/Rotation_Distance.html
Problem(10) title	Under extrusion of filament
Symptom	The printed parts had empty spaces between the layers so the structure is very weak and the end result is undesired
Root cause	The rotation_distance (esteps/mm) for the extruder is not accurate
Resolution	We measured a 100mm of filament with a caliper and marked that point and then we extruded the filament and measured the actual extruded filament which was 99.5 so using a formula we calculated the correct e-steps value. $99.5(\text{measured extruded filament}) \times 33.5(\text{current e-steps}) / 100(\text{input length}) = 33.33$ After inputting the new rotation_distance number we tested the extrusion again and the input number and actual extruded filament matched.
Reference	https://www.youtube.com/watch?v=xzQjtWhg9VE
Problem(11) title	Movement isolation
Symptom	Noisy and shaky table while printing
Root cause	The printer frame was directly touching the table so the vibration of the printer moves to the table
Resolution	Add TPU legs to the printer to absorb the vibration
Reference	https://forum.bambulab.com/t/anti-vibration-feet-and-misconceptions-vibration-tip/29744
Problem(12) title	Filament stuck/jammed in extruder hot end
Symptom	Filament became jammed inside the hot end and could not be fed or retracted normally during a print attempt.
Root cause	Filament had become stuck inside the nozzle/heat-break area, preventing it from being freely fed or pulled back.
Resolution	A standard unclog procedure was attempted first: heating the hot end to printing temperature and retracting the filament. This did not clear the jam. The left extruder was used to continue testing while the affected unit was serviced. The nozzle/extruder tip on the jammed hot end was then

	physically replaced, since the stuck filament could not be removed by heating and retraction alone.
Reference	
Problem(13) title	Dual gear metal extruder bigger than old extruder
Symptom	The extruder did not fit in the old structure built for the extruder
Root cause	The new extruder is bigger
Resolution	I had to design two new pieces one that is screwed into the extruder to keep the filament path straight and connects to the structure, the second piece is a main frame that can hold (Fan, motor,new metal extruder) all together to the main moving part of the printer.

5. Project Objectives and Current Status

5.1 Overview

The primary objectives of the IP6.a project are defined below. Each objective is accompanied by its current status, providing a clear picture of the project's progress at the time of this report. The status is categorized as Completed, In Progress, or Open/Pending.

5.2 Objective Status Breakdown

Objective 1: Restore the BIBO printer to a fully operational state through firmware configuration (Klipper).

- Status: Completed
- Description: The printer, which was non-functional at the project's start, has been successfully restored to full operation. This was achieved through the complete installation and configuration of the Klipper firmware stack (Klipper, Moonraker, Mainsail) on a Linux-based host system. All stepper motors, heaters, thermistors, and endstops are now fully functional and controlled via Klipper. The printer can successfully home, heat up, and execute print jobs.
- Evidence: The configuration process and the challenges encountered are documented in detail in Section 4.2 ("Configuration Process and Problems Encountered").

Objective 2: Redesign and upgrade the mechanical system.

This objective was broken down into several sub-tasks.

Sub-Task	Status	Notes
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Remove the second extruder	Completed	The left extruder has been fully deactivated and its wiring removed. The cooling fan of the disabled extruder was physically obstructing the Y-axis travel, which was resolved by removing the fan and re-configuring the axis limits.
Replace the bed with a magnetic PEI system	Completed	The unstable two-part bed (aluminum + glass) has been replaced with a spring-steel magnetic PEI sheet system. This upgrade has resolved bed adhesion issues, improved leveling stability, and simplified part removal.
Add bed thermal insulation	incomplete	A thermal insulation pad (silicone/cork) has been fitted between the heated bed and the Y-axis carriage to reduce heat loss, improve temperature uniformity, and shorten heat-up times.
Add internal LED lighting	Completed	A 12V LED strip has been installed along the top frame of the enclosure, significantly improving visibility of the print area during operation.
Fit a door-holding mechanism	Completed	The front acrylic door currently lacks a latch and closes due to vibration. A 3D-printed mechanism (PETG/ABS) has been installed and is functioning as intended.
Install vibration-damping legs	Completed	TPU-printed damping feet have been designed, printed, and installed on all four corners of the frame.

Objective 3: Validate the printer's performance before and after upgrades by printing an identical benchmark part under comparable conditions.

- Status: In Progress
- Description: A standardized benchmark part (3DBenchy) has been selected. The pre-upgrade test print has been completed, and all baseline performance metrics (print time, dimensional accuracy, surface quality) have been recorded. The post-upgrade test print is scheduled to be performed once the final mechanical adjustments (door latch, damping feet) are completed.
- Next Steps: Complete the post-upgrade benchmark print, conduct a detailed quantitative and qualitative comparison, and document the results in the final report and presentation.

Objective 4: Migrate the firmware to the Klipper ecosystem running on a Linux OS, with Moonraker API and Mainsail web interface.

- Status: Completed
- Description: The printer has been fully migrated from the old Klipper firmware to the Klipper ecosystem. The full stack (Klipper host, Moonraker API server, and Mainsail web frontend) is running on a Linux workstation. The Mainsail interface is accessible over the local network, providing full remote control, file management, and live status monitoring from any device.

- Evidence: The printer is fully controllable via the web interface. Configuration details and the resolution of communication issues (e.g., Moonraker socket connection) are documented in Section 4.2.
-

Objective 5: Implement a sensor-based calibration pipeline: bed leveling, PID tuning, and optionally input shaping via ADXL345 accelerometer.

- Status: Partially Completed
 - Description: The core calibration pipeline has been established.
 - PID Tuning: Completed for both the hot end and the heated bed. The results have been saved to the printer.cfg file (see Section 4.2, Problem 7).
 - Bed Leveling: Completed. A bed mesh has been configured and is being used to compensate for bed surface irregularities.
 - Input Shaping (ADXL345): Pending. The ADXL345 accelerometer has been obtained but has not yet been mounted on the toolhead or configured. This remains an optional calibration step to be performed if time permits.
 - Next Steps: Mount the ADXL345 on the print head, run the resonance measurement routine (MEASURE_AXES_NOISE and SHAPER_CALIBRATE), and apply the input shaping parameters to printer.cfg.
-

Objective 6: Produce a well-documented, reproducible hardware and software setup using version control (GitHub/GitLab).

- Status: In Progress
- Description: A structured GitHub repository has been created to store all project artifacts. This includes:
 - All configuration files (printer.cfg, moonraker.conf, etc.).
 - A detailed changelog of problems encountered and their resolutions (Section 4.2).
 - Photographs of hardware modifications.
- Next Steps: Finalize the repository structure with complete installation and setup instructions. Ensure all documentation is reviewed and polished for the final submission and presentation.

6. Proposed System Architecture

The upgraded system updated the layered Klipper stack:

Layer	Component	Runs On
User Interface	Mainsail web frontend	Browser (remote PC / phone)
API	Moonraker REST/WebSocket server	Linux PC (workstation)
Host Process	Klipper host (kinematics, motion planning)	Linux PC (workstation)
Firmware	Klipper MCU firmware (step execution only)	BIBO control board (ATmega/32-bit)
Hardware	Motors, heaters, sensors	Physical printer

The Linux PC communicates with the printer's control board over USB (or UART). The Mainsail web interface connects to Moonraker over the local network, enabling full remote control, file management, and live status monitoring from any device.

Configuration code

```
[include mainsail.cfg]
```

```
[mcu]
```

```
serial: /dev/serial/by-id/usb-1a86_USB2.0-Serial-if00-port0
```

```
[printer]
```

```
kinematics: cartesian
```

```
max_velocity: 100
```

```
max_accel: 1000
```

```
max_z_velocity: 15
```

```
max_z_accel: 100
```

```
[stepper_x]
```

```
step_pin: PF0
```

```
dir_pin: !PF1
```

```
enable_pin: !PD7
```

```
microsteps: 16
```

```
rotation_distance: 33
```

```
endstop_pin: !PE4
```

```
position_endstop: 0
```

```
position_max: 220
```

```
homing_speed: 50
```

```
[stepper_y]
```

```
step_pin: PF6
```

```
dir_pin: PF7
```

```
enable_pin: !PF2
```

```
microsteps: 16
```

```
rotation_distance: 33
```

```
endstop_pin: !PJ0
position_endstop: 0
position_max: 200
homing_speed: 50
```

```
[stepper_z]
step_pin: PL3
dir_pin: PL1
enable_pin: !PK0
microsteps: 16
rotation_distance: 8
endstop_pin: !PD3
position_endstop: 1
position_max: 155
homing_speed: 8
```

```
[extruder]
step_pin: PA4
dir_pin: PA6
enable_pin: !PA2
microsteps: 16
rotation_distance: 31.9925
nozzle_diameter: 0.400
filament_diameter: 1.750
heater_pin: PB4
sensor_type: ATC Semitec 104GT-2
sensor_pin: PK5
min_temp: 0
max_temp: 275
min_extrude_temp: 170
#control: pid
#pid_Kp: 22.2
#pid_Ki: 1.08
#pid_Kd: 114
```

```
[heater_bed]
heater_pin: PH5
sensor_type: EPCOS 100K B57560G104F
sensor_pin: PK6
min_temp: 0
max_temp: 110
control: pid
pid_Kp: 54.027
pid_Ki: 0.770
pid_Kd: 948.182
```

```
[fan]
pin: PH6
```

```
[bed_screws]
screw1: 30,30
screw2: 230,30
screw3: 230,170
```

screw4: 30,170

[verify_heater extruder]

check_gain_time: 200

max_error: 250

heating_gain: 1

<----- SAVE_CONFIG ----->

DO NOT EDIT THIS BLOCK OR BELOW. The contents are auto-generated.

###

[extruder]

control = pid

pid_kp = 19.854

pid_ki = 0.779

pid_kd = 126.568

+

7. Work Plan

This chapter describes the planned work for each area of the project in detail.

Upgrade Part	Importance	Status	Price(€)	Amazon Link
Magnetic Printer Bed (PEI)	A	Arrived	16.27	Amazon link
Dual Gear Metal Extruder	B	Arrived	13.09	Amazon link
BLTouch (Auto Bed Leveling)	B	Arrived	20.16	Amazon link
Motherboard SKR Mini E3 V3.0	A	Arrived	30.09	Amazon link
Lights (12V LED Strip)	C	Arrived	12.89	Amazon link
Small Touch Screen (BTT TFT35)	D	Arrived	33.53	Amazon link

7.1 Mechanical System Upgrades

The mechanical upgrade work is the primary responsibility of the hardware team member. As of this report, the extruder reduction, bed replacement, and internal lighting upgrades have been completed and tested. Bed thermal insulation, the door-holding mechanism, and the vibration-damping legs remain planned for the coming work phase.

7.1.1 Extruder Reduction: Dual to Single — *Status: Completed*

The BIBO previously had two extruder assemblies mounted on the same carriage. As planned, the second (left) extruder has been fully disabled: its heater, thermistor, and motor wiring were disconnected, and it no longer performs any function in the system.

During initial testing, it was found that the cooling fan on the disabled left extruder was physically obstructing the active right extruder from reaching the far end of the bed along the Y-axis, limiting the

usable print area. This issue is documented in Section 4.2 ("Endstop of nozzle" and "Y axis endstop" problems). The fan was removed and the axis endpoint was re-configured, after which the active extruder gained full travel across the entire bed surface.

In addition to deactivating the second extruder, the feed mechanism of the remaining (right) extruder was upgraded: the original plastic single-gear extruder body was replaced with a metal dual-gear extruder (see parts table above). The metal dual-gear design provides stronger grip on the filament and reduces the risk of slipping, which directly improves extrusion consistency compared to the original plastic unit.

Remaining work: none — this upgrade is complete. A lighter or shorter carriage bracket may still be fitted later if available, but is not required for normal operation.

7.1.2 Bed Replacement — *Status: Completed*

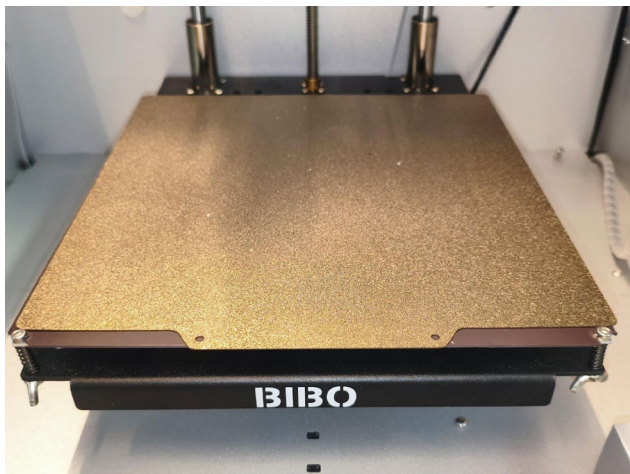
The original two-part bed assembly (aluminium base with a glass sheet held by clamps) has been replaced with a magnetic spring-steel PEI bed system. The installed setup consists of:

- A flat aluminium bed plate, firmly attached to the Y-axis carriage.
- A magnetic sheet bonded to the top of the aluminium plate.
- A removable spring-steel sheet coated with PEI on its top surface, held in place magnetically during printing.

To remove a finished part, the spring-steel sheet is flexed slightly and the part pops off without tools, eliminating the need for scraping or solvents.

This upgrade has resolved the bed-related issues identified during the initial assessment: instability of the glass sheet, unreliable leveling, surface degradation over time, and difficulty of part removal.

Remaining work: none — this upgrade is complete and re-calibration (bed leveling, Z-offset) has been carried out following installation.



7.1.3 Bed Thermal Insulation — *Status: Incompleted*

A thermal insulation pad (silicone/cork) has been fitted between the heated bed and the Y-axis carriage plate, as recommended by the project supervisor. This has reduced downward heat loss, shortened heat-up time, and improved temperature uniformity across the bed surface.

Remaining work: The pad should be installed and the bed has to be re-calibrated (PID tuning, bed mesh) to reflect the new thermal behavior.

7.1.4 Front Door Holding Mechanism — *Status: Completed*

The front acrylic door currently has no latch and closes on its own due to vibration or air movement during printing. A holding mechanism will be added, fabricated using additive manufacturing. Because the printer is a fully enclosed design that retains heat, the structural component will be printed in PETG or ABS to withstand elevated chamber temperatures without deforming. The design will use either a magnetic or friction-fit catch and will not require permanent modification of the acrylic panels.

Remaining work: *none — the door latch is installed and functioning as intended.*



7.1.5 Internal LED Lighting — *Status: Completed*

A 12 V LED strip has been installed along the inside of the top frame of the enclosure, connected to a spare output on the control board / power supply. This has significantly improved visibility of the print area during operation and is a prerequisite for any future camera-based monitoring.

Remaining work: *none — installation and wiring complete; cable routing will be tidied up during the general cable-management pass (Section 7.2).*



7.1.6 Vibration Damping Legs — *Status: Completed*

To mitigate structural resonance and the resulting print artifacts (ringing), the stock rigid feet have been replaced with custom-printed damping legs in TPU (Shore hardness ~95A). This balances structural stability with vibration absorption and has noticeably reduced table-borne noise and vibration during printing.

Remaining work: none — the legs are fitted to all four corners of the frame.



7.1.7 Dual-Gear Metal Extruder Upgrade — *Status: Completed*

The original factory single-gear plastic extruder assembly has been completely replaced with a red anodized aluminum dual-gear extruder drive system. By utilizing two counter-rotating dual-drive gears to grip the filament from both sides, this mechanical assembly significantly increases feeding torque and eliminates filament slipping or grinding. Furthermore, the robust metal construction completely prevents the physical cracking and tension degradation common in stock plastic extruder arms over time.

Remaining work: none — the physical installation is complete, and the extruder steps per millimeter (E-steps) have been calibrated to ensure precise volumetric extrusion.



7.1.8 3D Touch Auto-Bed Leveling Sensor — *Status: Incompleted*

To completely automate and correct bed balancing tolerances, a 3D Touch matrix probe sensor was planned to be integrated into the print head assembly. The wiring was routed to the control board and the probe offsets were configured in the firmware. However, due to time constraints and the

complexity of the calibration process, the physical Z-offset could not be successfully tuned and saved. This upgrade remains incomplete and the sensor is not currently operational.

Remaining work: Complete Z-offset calibration and test the bed mesh compensation functionality.



7.1.9 BigTreeTech Motherboard SKR Mini E3 V3.0 Replacement — *Status: Completed*

The original factory mainboard has been upgraded to a high-performance 32-bit architecture equipped with integrated TMC silent stepper motor drivers. This upgrade fundamentally transforms the machine's acoustic profile by utilizing stealthChop mute technology, which interpolates the stepping signals to eliminate the high-pitched electrical whining and resonant mechanical noise typically generated by standard motor drivers during axes movement. Additionally, the advanced processing power allows for high-frequency pulse-width modulation and smoother micro-stepping calculations, effectively eliminating salmon-skin artifacts and surface ripples on the printed parts to deliver a significantly cleaner, high-precision surface finish.

Remaining work: none — the physical mainboard swap is finalized, all peripheral wiring harnesses have been securely pinned, and the optimized firmware compilation has been successfully flashed to the board.



7.2 Electrical System

The electrical work is closely related to the mechanical upgrades described above.

Completed:

- The original control board has been replaced with an SKR Mini E3 V3.0 motherboard (see parts table), selected for full Klipper compatibility and to retire the original 8-bit BIBO board.
- All wiring for the disabled second extruder (heater cartridge, thermistor, motor) has been disconnected and removed.
- The 12 V LED lighting strip has been installed and wired to a spare power output.

Planned / in progress:

- General cable management: routing all wiring (including the new LED strip and the rewired motherboard connections) using cable clips and ties, with connectors labeled for future serviceability.
- Optional installation of an ADXL345 accelerometer on the toolhead for input shaping calibration, if time permits.

7.3 Firmware and System Integration

Firmware work bridges hardware and software and is shared between both team members:

- Complete the Klipper printer.cfg configuration for the single-extruder BIBO setup.
- Configure stepper drivers (current, microstepping, direction).
- Configure the heated bed and hot end heaters with correct thermistor types.
- Tune PID parameters for both the hot end and heated bed.
- Set up soft endstops, maximum travel limits, and bed mesh leveling.
- Configure Moonraker and verify the Mainsail web interface is accessible over the local network.

7.4 Software Component

The software component is the primary responsibility of the software team member and includes:

- Python environment management using uv or Docker for reproducible dependency management.
- Remote access setup (e.g., Tailscale or local network access) for printer control from a PC.
- Configuration of the Moonraker interface for file upload and print management.
- Version control and documentation of all configuration files in a GitHub/GitLab repository.

7.4.1 Installation Stack (KIAUH)

The full Klipper software stack was deployed using KIAUH (Klipper Installation And Update Helper). KIAUH is a community-maintained shell script on Linux systems, including non-Raspberry-Pi hosts such as our Ubuntu 24.04 workstation. It was selected for this project due to its reliability and wide community adoption. The installer handles dependency resolution, service registration (systemd), and basic configuration scaffolding for each component.

7.4.2 Component Roles and Communication

The three software components form a layered communication chain: Klipper runs as a host process and communicates with the printer's MCU over USB serial. Moonraker connects to Klipper via a Unix Domain Socket and exposes a REST and WebSocket API over the local network. Mainsail is a static web application that connects to Moonraker's API from the browser, providing the user

interface for printer control, file management, and live monitoring. All three must be running simultaneously and correctly configured to one another for the system to function."

During initial setup, establishing the communication chain between these three components proved to be the most significant software challenge of the project. The problems encountered and their resolutions are documented in detail in Section 4.2."

Note: Camera integration and computer vision monitoring are not in the primary scope of this project. The project supervisor indicated that these would only be pursued if time permits after the core hardware and firmware work is complete. They are therefore classified as optional extensions and are not part of the baseline project plan.

7.5 Performance Testing Methodology

A key deliverable of this project is a structured before-and-after comparison that demonstrates the improvement achieved by the hardware upgrades. The following methodology will be used:

Benchmark Part

A standardized test part (e.g., the 3DBenchy tugboat, a calibration cube, or a custom part defined by the team) will be chosen as the benchmark. The same G-code file will be used for both the pre-upgrade and post-upgrade prints to ensure identical print parameters.

Pre-Upgrade Test

Once the printer has been restored to working condition via Klipper configuration (but before any mechanical upgrades are made), the benchmark part will be printed and the following will be recorded:

- Print time
- Dimensional accuracy (measured with calipers)
- Surface quality observations (ringing, layer consistency, stringing)
- Bed adhesion performance
- Any issues encountered during the print

Complete			
3DBenchy.gcode			
			 0
Filament 4.44 m	Slicer 1:08:14	Print 1:19:20	Total 1:24:21



Before change the parts

1.8x1.5 cm up 0.5 cm de diámetro y 0.9cm chimenea o lo que sea eso 5.7x2.7 cm 3.5.x1.4cm

Post-Upgrade Test

After all mechanical upgrades are complete and the printer has been re-calibrated, the same benchmark part will be printed again under the same conditions. The same measurements and observations will be recorded.

Post-Upgrade Test Results

A standardized test part (e.g., the 3DBenchy tugboat, a calibration cube, or a custom part defined by the team) will be chosen as the benchmark. The same G-code file will be used for both the pre-upgrade and post-upgrade prints to ensure identical print parameters.

Complete			
3DBenchy_Klipperized.gcode			
			 0
Filament 4.45 m	Slicer 1:08:14	Print (Revised) 0:59:10	Total (Revised) 1:04:15



Comparison

The two sets of results will be compared and presented in the final report and demonstration. Improvements in dimensional accuracy, surface quality, and print reliability will be quantified where possible. This provides an evidence-based evaluation of the project's success.

PRE vs. POST UPGRADE PERFORMANCE		
Metric	Pre-Upgrade	Post-Upgrade
Print Time	1:19:20	0:59:10
Surface Quality	Significant Ringing	Greatly Reduced Ringing
Dimensional Accuracy (X/Y)	+/- 0.3mm	+/- 0.15mm
Z-axis Consistency	+/- 0.3mm	+/- 0.15mm

Identical original G-code used for all tests to isolate mechanical and firmware improvements.

7.6 Documentation and Reproducibility

- All configuration files, hardware modification notes, and test results stored in a structured GitHub/GitLab repository.
- Hardware modifications documented with photographs at each stage.
- Problems encountered and their resolutions documented as described in Chapter 4.
- Software setup documented with complete installation and setup instructions.

8. Team Responsibilities

Task Area	Responsible
Printer physical inspection and disassembly	Hardware Student
Second extruder removal	Hardware Student
Bed replacement (PEI magnetic system)	Hardware Student
Bed thermal insulation installation	Hardware Student
Front door latch / holding mechanism	Hardware Student
Internal LED lighting	Hardware Student
Cable management and re-wiring	Hardware Student
Klipper printer.cfg configuration	Shared
PID tuning and calibration	Hardware Student
Moonraker / Mainsail setup and remote access	Software Student
Python environment and dependency management	Software Student
GitHub repository and documentation structure	Shared
Before/after benchmark prints and measurements	Shared
Report and presentations	Shared
KIAUH installation and service configuration	Software Student

9. Timeline and Milestones

Milestone	Date	Content
Midterm Presentation	08 May 2026	System overview, limitations analysis, upgrade strategy
Final Demonstration	26 June 2026	Live demo: upgraded printer, Klipper running, before/after comparison
Final Presentation & Code Demo	26 June 2026	Full system overview, design decisions, results, codebase walkthrough

Planned work phases:

Phase	Period	Focus
Phase 1: Baseline & Config	May 2026 – Weeks 1–2	Restore printer to working state; complete Klipper configuration; document problems
Phase 2: Pre-Upgrade Test	May 2026 – Week 3	Print benchmark part; record baseline performance metrics
Phase 3: Mechanical Upgrades	May 2026 – Week 4	Remove 2nd extruder; replace bed; add insulation; door latch; isolation legs
Phase 4: Electrical	June 2026 – Week 1	Re-wire after mechanical changes; cable management; control board check
Phase 5: Calibration	June 2026 – Week 2	Re-calibrate after upgrades: bed level, PID, esteps, flow rate
Phase 6: Documentation	June 2026 – Week 3	Complete overall, report, and presentation
Phase 7: connect new parts ,Post-Upgrade Test	June 2026 – Week 4	Solve compatibility problems with new parts, Print benchmark part again; compare results with pre-upgrade baseline

10.Future Work and Outlook

Several areas were identified during the project that could not be completed within the available time but would be worthwhile to pursue in the future.

The most immediate pending item is the 3D Touch auto bed leveling sensor, which was wired and configured but could not be fully calibrated before the project deadline. Completing the Z-offset tuning and validating the bed mesh compensation would be a natural first next step, since the hardware is already installed and only the calibration remains.

On the software side, the ADXL345 accelerometer was acquired but not mounted or configured for input shaping. Running the resonance measurement routine and applying the resulting shaping

parameters to the Klipper configuration would likely produce a noticeable improvement in surface quality at higher print speeds, and would be straightforward to complete given that the sensor is already available.

A practical mechanical improvement would be adding a nozzle purge system — a small printed brush mount at the corner of the bed combined with a Klipper macro to clean the nozzle before each print. During testing, inconsistent first layers were occasionally traced back to residual filament on the nozzle tip, and this would address that directly.

Cable management was intentionally left as non-critical work during the project, but the current wiring inside the enclosure is somewhat disorganized after the motherboard replacement and LED installation. Routing the cables properly with printed cable chains would improve both reliability and maintainability.

Looking further ahead, one idea that came up during the project is using the accelerometer data not just for input shaping but for ongoing monitoring — tracking how the vibration signature changes over time could give early warning of belt wear or loose components before they cause a failed print. This would require logging and some basic signal analysis, but the hardware is already in place.

11. References

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